Metaphors in co-designing a digital healthcare platform: the case of the Brain House for the Health Village

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Abstract

This empirical and method study explored the co-design processes of online public health care services for a national digital platform in Finland. The aim of the research was to understand how multi-stakeholders co-designed content for online services of a digital platform, and how metaphors impacted collaborative sensemaking processes and action. Our research design applied qualitative interviews, observations, and two workshops with Collaborative Story Craft method to explore the narratives and the sensemaking processes of multi-stakeholders involved in the co-design of the Brain House for the Health Village. The Brain House represents the specialization of neurology and is one of 32 houses for the Health Village, which is nationally accessible and coordinated by the five main university hospitals. Through the interviews and the workshops, we argue that the metaphor of the "Brain House" of the "Health Village" revealed a metaphoric field that facilitated collective sensemaking and leveraged collaboration among multi-stakeholders and researchers themselves.

1. Introduction

Digital healthcare platforms fall under the category of e-health, which, broadly defined, is "the application of ICT [internet communication technology] to support the organization,

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management and delivery of healthcare" (Doolin, 2016: 352) by multiple and diverse users (Islind, 2019: 426).

Digital healthcare platforms are common tools used to support cooperative care between healthcare practitioners and the community they serve. Digital platforms as a mean to transform care processes are increasingly viewed to facilitate patient-centred healthcare (Islind *et al.*, 2019) and to collaborate across communities of practice at the organizational level (Struminger *et al.*, 2019). Governmental reforms have also been far-reaching with large investments in technology as well as Public-Private Partnership (PPP). The justifications for the PPPs and technological investments are to save money while fostering competition and choice to stimulate innovation and improvement (Bishop and Waring. 2016: 459) through digitalization reforms (Denis *et al.*, 2001).

However, despite large governmental investments and optimism into e-health like digital platforms, many scholars have argued that the benefits of e-health are more anticipated than realised in practice, and there is a lack of empirical evidence of its achievement (Black *et al.*, 2011; Goldzweige *et al.*, 2009; Lau *et al.*, 2010; Agarwal *et al.*, 2020). This suggests that implementing e-health in any meaningful way requires going beyond the technology and a commitment to understanding, instead, changes in the work processes, interactions, and behaviours of both practitioners and patients (Agarwal *et al.*, 2010; 2020; Blumenthal and Glaser, 2007; Pope *et al.*, 2013).

In this empirical article, we explore the co-design processes by multi-stakeholders for the Finnish "Health Village": a digital service platform developed with patients, healthcare professionals and patient organizations to deliver public health services. The aim of the Health Village is accessible care for the general public, digital care pathways for patients with specific diagnoses, digital guides and online training courses in eExpertise for professionals to improve working practices. Data is also accumulated through the Health Village for research, development, and quality improvement activities.

In four years (2016-2018), the Health Village scaled nationally, and the project received funding from the Finnish Ministry of Social Affairs and Health. The Health Village was organized in collaboration with the university hospitals of Helsinki, Kuopio, Turku, Tampere, and Oulu (Helsinki University Hospital, 2020).

In our research, we wanted to explore how the platform was co-designed by multi-stakeholders through interviews, observations and two workshops with those involved with the specific diagnoses of neurology.

2. Co-designing the "Brain House" for the "Health Village"

The metaphor "Health Village" was given by an external IT company in 2014 who created and designed the technical backend and frontend of the cloud service platform. The IT functional purpose of the Health Village was to provide a cloud service platform so that patients, healthcare professionals and patients organizations could co-design accessible public health care services. This co-design required collaboration and coordination across the five main hospitals, various advocacy and research organizations for patients, and reaching out to former patients with specific diagnoses (Helsinki University Hospital, 2020). In lieu of the metaphor Health Village, the co-design processes were framed in Finnish as "Talo" (House),

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meaning the stakeholders' activities were to co-design accessible diagnostic houses for the Health Village.

Between 2016-2018, the Finnish Ministry of Social Affairs and Health granted €6 million for the project "Public services will be digitised". The project involved all five major university hospitals which matched the Ministry's funding, bringing it to a total of €12 million. Over the two-year period, later extended to four years (2016-2020), the focus areas were (1) the production and implementation of online services such as offering information, advice, self-care, symptom navigators, digital treatment pathways, and tools for citizens, patients and professionals; (2) the creation of an innovation farm which included workshops, piloting, artificial intelligence, research and the researcher's tools; and (3) the development of services and changes in operation such as development model, developer network and centres of expertise (terveyskylä.fi).

Governmental funded projects such as "Public services will be digitalised", which involve multiple stakeholders, are not new trends as public organizations have been researched as pluralistic for decades (*e.g*. Denis *et al*., 2001). In addition, "Public services will be digitalised" further illustrates decades-old development shifts which emphasize experiences, values, and quality of life of citizens and their participation in public organizations (Argwal, 2020).

1.1. Methods to explore co-design of the Brain House for the Health Village

Co-designed projects of health care services by providers, patients, and community members are common, government funded practices in democratic welfare states. The digitalization of these services creates no exception as e-health is considered a strategic way to address decades-long challenges such as accessibility, cost efficiency, retaining and the management of highly skilled professionals, and innovation through technology (Argwal, *et al*., 2010; Argwal, 2020).

Research of co-design in the healthcare context requires a qualitative, participatory research design that is versatile, adaptable, and people focused (Groeneveld *et al*., 2019: 314). It also requires gaining access, managing relationships, building understanding, and communicating value in a highly regulated professional organization (Groeneveld *et al*., 2019: 316; Denis *et al*., 2001).

From the beginning of our research, the technical purpose and the metaphor of the Health Village facilitated communicating and building understanding of the digital platform as a metaphoric boundary object (Koskinen, 2005). The semi-open metaphoric boundary object provided enough space, flexibility, and possibility for our request to participate, observe, interview, and conduct two final workshops over a 12-months period (2018-2019).

The theoretical premise of our research design was Collaborative Storytelling. Collaborative Storytelling advocates for space and time to reflect on social identities and embodied, lived experiences through stories in relation to an activity (Cleland Silva and Fonseca Silva, 2022: 1). We wanted to explore the co-design processes through interviews with the multi-stakeholders, participate in formal meetings as well as informal discussions about the Health Village over an 18-month period.

Indeed, at times, the research design felt *ad hoc* as we committed to preserving space and time for reflections by the multi-stakeholders which included our research team. In addition, members of our research team came from different social worlds (Strauss, 1978). As the only

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non-Finnish citizen, the main researcher for the method design and first author had limited ability to communicate and understand Finnish. All the others, whether from the Health Village or from our research team were citizens of Finland whose knowledge level of the Finnish language was either native or advanced. That said, most of the stakeholders did welcome interviews in English and the two final workshops were held predominantly in English. The Finnish speaking research members conducted the interviews requested to be in Finnish and translated the professional transcriptions of the Finnish interviews (see Appendix 1). The research team subsequently held frequent meetings in person to discuss the interviews which were framed with an interview guide (see Appendix 2). In addition, two Finnish speaking research members assisted the non-Finnish researcher to translate or communicate nuances of the strategic meetings under observation. When the non-Finnish researcher was alone, the stakeholders of the Health Village consistently asked and accommodated if the language became a barrier.

All the interviews and observed meetings, formal and informal, were audio recorded. The formal meetings that we were able to schedule in advance were also video recorded. In total, 16 hours of meetings, including conversations in a hallway and a cafeteria, were audio recorded and transcribed. Ten hours of observations were video recorded. The final two workshops were both video and audio recorded, eight hours' worth of recording whose audio was subsequently transcribed.

The intention for the audio and video recordings was to document the group's sensemaking of "what's the story" (Weick *et al.*, 2005) of the Brain House in the Health Village. We aspired to explore sensemaking's intertwined facets of noticing, meaning making, and acting (Maitlis and Christianson, 2014) through the co-design activities. The Health Village and the Brain House strategic aims of accessible care made it easier for us to explore the co-design process of the multi-stakeholders.

In addition, the Brain House as a metaphoric boundary object facilitated the research process as an ongoing and unfolding sensemaking process of metaphor (Langley, 2021). The collective sensemaking of "what's the story?" became "what's the metaphor?" as a chosen and conceptual frame of scientific inquiry (Cleland Silva, 2022: 220).

1.2. Gaining trust of co-design stakeholders to explore sensemaking

As one of our research members was employed by the neurological unit at the Helsinki University Hospital (HUS) for the Health Village project, our first research meeting was with the project leader who is a neurologist.

During our first meeting, we were informed that the neurological unit at HUS as well as four other research hospitals were co-designing online information for the "Brain House" (*Aivotalo* in Finnish) as well as digital pathways. The "Brain House" had eight specialized groups, and the leader welcomed us to research the headache and stroke groups since both groups were in the early stages of the co-design process (research diary entry May 2018).

Immediately, we emailed the groups' leaders to request research access, and both were open to our suggestions to use video and audio recording equipment in meetings and interviews with stakeholders from the headache and stroke groups.

Over a month (the end of May until the end of June 2018), our research team interviewed 18 stakeholders from the stroke and headache groups as well as the Health Village. From July

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2018 to April 2019, the primary author continued the observations alone, and in April and May 2019, the primary author designed and facilitated the Collaborative Story Craft workshops with the second and the third authors.

1.3. Managing relationships and communication of multiple stakeholders in health care

In the intervention method of Collaborative Story Craft, researchers serve as Story Mediators to help externalize various voices into a holistic interpretation of a situated activity (Cleland Silva and Fonseca Silva, 2019; 2022).

The intention of this method is to facilitate individual interpretations in organizational settings to identify themes and metaphors that unite the actors. Once themes are identified, the Story Mediator deepens the significance of the metaphor in relation to the activity through a collective effort with practitioners. This is usually in the form of a workshop.

The primary author acted as a Story Mediator and designed the research, managed the relationships with stakeholders, and led the final Collaborative Story Craft workshops. In practice, holding space, active listening, and mediating the stories about the co-design of the Brain House and the Health Village required a tailored design of the research context and an awareness of the stakeholders' social worlds, including all the researchers (Cleland Silva and Fonseca Silva, 2022: 12-17). In the context of a Finnish hospital, this meant flexibility of time schedules and facilitating language access and use for stakeholders to feel psychologically and physically safe (*e.g.*, Kahn 1990) to express themselves. It also required our research team's collective movement between the centre and the periphery of the stories (Choudhury, 2015: 82-84) or as Patricia Hill Collins referred to "learning from the outsider within" (1986).

3. Metaphors in co-designing a digital healthcare platform: Making sense through Collaborative Story Craft

1.4. Phase 1: Interviews with the Brain House stakeholders

Interviews with the stakeholders centred on the co-design processes of online information and digital pathways for the neurological diagnostics of headache and stroke. In principle, the headache and stroke groups were tasked to co-design self-care platforms to deal with patients' needs and enable them to have access to a channel to share information with the practitioners. For instance, within the headache digital pathway, patients were expected to keep a daily digital diary in which practitioners interacted with them over an eight-month period.

Although interviews came with semi-structured questions (see Appendix 2) to help each researcher facilitate the story exchange about the Brain House and Health Village, researchers expressed an interest in and were open to hearing about experiences with the situated activity. This allowed them to document the processes of noticing and meaning making through narratives as a "process as withness" (Shotter, 2006).

Narration is defined as the signifier, stories are signified by the narrative (Snaevarr, 2010: 168) and both narrative and story are impacted by the social environment. Each interview, observation, and final workshops were documented at one of two sites which belong to HUS:

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the neurological unit in the hospital and a research building, located 300 meters from the hospital. These sites both have cafeterias and staff kitchens. At the research building, there are seminar rooms for meetings. At the hospital, most of the practitioners treated patients from the neurological units and meetings were mostly held in the staff kitchen.

1.5. Phase 2: Validation of stories around the activity

In practice, Story Mediation is used to externalize and identify metaphors and themes that unite the storytellers of a situated activity (Cleland Silva and Fonseca Silva, 2022: 19). Identifying the metaphors and themes is usually crafted with another Story Mediator through a process of "offloading" to validate the story and craft "mirror material" for the original storyteller. As the word suggests, "mirror material" is used to reflect the Story Mediator's interpretation of the data back to the storyteller to support the sensemaking process.

In the case of the Health Village and Brain House, the metaphor preceded the co-design processes. For this reason, validation of the narratives was not done in the original form of Collaborative Story Craft, which is to hold another one-on-one session with the storyteller to mirror back their stories and sensemaking. Instead, throughout the year, the primary researcher, who is a Story Mediator, was in continuous dialogue with the stakeholders, always bringing back her materials and sharing her findings through diary entries, PowerPoint presentations, photographs, and maps so that stakeholders could validate, correct, or complement the mirror material. The dialogue between the stakeholders and the researcher revealed common concerns related to time, roles, responsibilities, and expectations of their work in relation to the co-design. Most of the stakeholders articulated that this novel and innovative activity of co-design had much potential but that they were constrained by time and resources allocated by their superiors. These constraints were believed to originate from the pioneer nature of the new tool, but also from the demanding reality of health care practitioners' work.

In addition to complementing the materials, the health care practitioners mostly approved the existing interpretation materialized by the researcher. The validation of the mirror materials allowed the identification of themes and metaphors that were relevant to tailor the upcoming workshops.

This type of dialogic and validation research has been referred to by other scholars as "situated dialogic action research" (Shotter, 2006), "dialogical mediated inquiry" (Lorino *et al.*, 2011), "collaborative action research" (Fachin and Langley 2017: 320) and "engaged scholarship" (Van de Ven 2007).

1.6. Phase 3: Final workshops: Building understanding and communicating value of the metaphor

After a year of research, two workshops were designed by the Story Mediator for those interested and from within the headache and stroke groups in the Brain House. This meant that the workshops were open to stakeholders interested and motivated to participate. The workshops were held on separate dates to accommodate the work schedules of the healthcare practitioners.

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The first workshop, conducted with the headache group, consisted of the Brain House leader, (a neurologist), the headache's group leader (a neurologist), and two hospital administrators. The second workshop consisted of stakeholders from the stroke group and involved the stroke's group leader (a neurologist), a stroke nurse, a patient, and five members of a Finnish brain health association. The Brain House leader participated again in this workshop.

The first workshop lasted 4 hours and the second, lasted two hours. Each workshop started with an overview of the Collaborative Story Craft methodology (Figure 1).

Figure 1



Figure 1: Cycle of Collaborative Story Craft presented at the workshop.

During each workshop, the Story Mediator and the research team member documented the event with notes. The Story Mediator initiated the story exchange by first exploring the experiences of the multi-stakeholders with the digital platform by posing these questions:

- How have your experiences been with working on the Health Village for the headache group or stroke group?
- What were the main achievements?
- What were the main challenges?
- What were the main lessons learned?
- What are you grateful for?

This initial exercise allowed stakeholders not only to express their understanding of the activity, but also to vocalize their role and witness how their co-designers interpreted the situated activity themselves. This process was brief as the questions served to provoke collective sensemaking to initiate the intervention.

The workshop then turned to the multi-stakeholders' stories as individuals by posing these questions:

- How and why did you become a health care practitioner?

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- When did you start working with HUS [the hospital in the research]?
- How would you describe your role at HUS?

These questions are part of an activity introduced by the Story Mediator called "Nurturing the Roots" (Cleland Silva and Fonseca Silva, 2022: 62-63). The metaphor was used to reconnect with the origins and reasons that motivated stakeholders to co-design the Brain House. The reasoning behind these questions is for the stakeholders to reflect on their professional identity (individually and collectively) in relation to their work environment and each other.

By answering the questions, each participant had an opportunity to share their story in relation to the questions while the other stakeholders actively listened by taking notes. Their notetaking followed some guidelines: "how does your colleague's story relate to yours? " and "what calls your attention to the story (something that you did not know before)?".

After each of the stakeholders has shared their story, their colleagues vocalized their feedback based on their notes.

Once this sharing activity was completed, the workshop turned to the metaphors of the Health Village and the Brain House used in the platform itself. To explore stakeholders' understanding of the activity, two images from the platform were used as a reference to situate and visualize the metaphors. The first included the representation of the Health Village (Figure 2), and the second was the image associated with the Brain House (Figure 3).



Figure 2: Front page of the Health Village digital platform.

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Figure 3

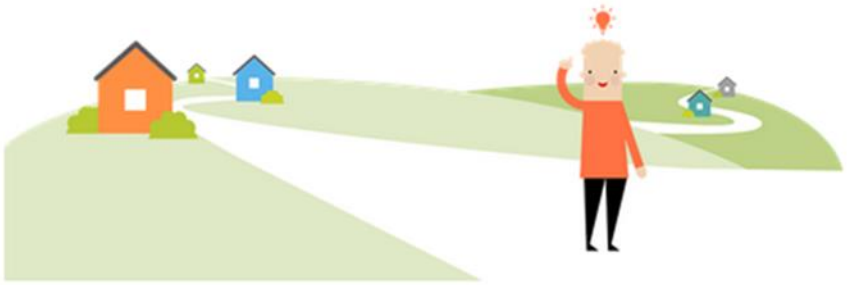


Figure 3: Avatar for the Brain House

In relation to the first image, the researcher posed these questions: how would you describe this image? What does the word Health Village mean to you? For the second image, the researcher asked participants to describe it and position it within the Health Village.

After this exercise, the Story Mediator introduced another of the method's metaphors called Metaphorical Tourism (Cleland Silva and Fonseca Silva 2019; Cleland Silva, 2022). This metaphor was for stakeholders to collectively envision a future story regarding the Health Village.

The prompting statements and questions on the future of the Brain House in the Health Village were:

- Your role and what you need to fulfil that role
- What is your destination / where do you want to go?
- Who are you travelling with?
- How should the responsibilities be divided?
- The map: What type of rules and guidelines should be put in place?
- What type of tools or resources do you need?
- Who is part of your community?
- Think about the landscape that surrounds the journey. What do you see?

Participants were asked to reflect on these and take notes. Subsequently, stakeholders were asked to use drawings that would reflect how they envisioned the Health Village. Once these were materialized, each stakeholder vocalized their ideas and expectations for the future of the Brain House and Health Village.

4. What's the Story of the Brain House? Collective Sensemaking through Collaborative Story Craft

1.7. Noticing the Brain House through interviews

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Scholars have argued that technology such as a digital platform functions as a boundary object bridging the gap between diverse communities of practice and social worlds (Bowker and Star, 1999; Star and Grisemar, 1989). Interviewing stakeholders about co-designing the Brain House for the Health Village helped us as researchers to identify the stakeholders and communities of practice involved.

During the interviews, we identified various communities of practice and social worlds such as the specialized diagnostic healthcare practitioners, IT developers, business managers, and the headache and stroke patients. There were also implied stakeholders such as families, citizens, future users, government, hospital administrators, and leaders as well as so-called "resources" to maintain the project (see Clarke *et al.,* 2017 for "implicit actors").

The Brain House, as a boundary object, became noticed and emerged through narrations of the co-design process as a tool for care reform: "The focus is more that we are reforming the care through that, digitalisation is just a tool, of reforming the mindset and everything that comes with (that). [0:31:24] And also that it's co-creative. [laughs] (Interview #18)". Collaboration was a central concern for all stakeholders. The project leader of the Brain House was in constant contact with both the headache and the stroke groups.

A psychiatrist also involved with the Village for another house voiced the importance of constant communication on the purpose and direction of the project: "the biggest contradiction is among our own people... there are different wants and needs and wishes" (Interviewee #11).

Through the interviews of co-design, noticing how health care practitioners have traditionally worked in "silos" (Interview #13) was emotionally narrated. Sensemaking processes are deeply emotional and narrating experiences reflects how a person feels rather than reporting facts (Greenhalgh, 1999). As one stakeholder admits of her past work focus (Interview #14): "I was an idiot that I was only focusing on my own profession and wasn't interested in what goes on in other professions in society".

Emotions were commonly narrated in the sensemaking processes. For instance, the leaders involved narrated their experiences as new with high expectations and pressure. As the leader of the stroke group said (Interview #1):

> and this is very new to me, and sometimes I think it's sort of, a bit unfair, that we are doing this, trying to develop this for the whole nation, since I think most of us really try to think… we hope that this will help really both individual patients and then the whole, (--) [0:00:51], national health, our public health. But we do this (even if), of course, university hospitals got a lot of money for this, so of course, this is expensive, and I understand that, a big portion of the money has to go to the data, sort of the data solutions and the IT solutions. But of course, also the content means something, and that is, we have to develop all the content with very little resources and training. (-) [0:01:33] everybody understands, and we try to do our best.

By narrating the process as "we try to do our best" also reflects a sense of vulnerability and openness to learning through the co-design process.

The sensemaking of the digital platform was retrospective but also prospective with the stakeholders' telling stories about their work's future (Interview #9): "Digitalization is the future, and a positive thing... Health care is lagging behind in development, but it is a positive thing that requires resources in the beginning and will later bring savings".

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Narrating on the future also revealed an "antenarrative" (Boje, 2011) or a "bet" that their current work will bring savings or pay off in the future.

Through dialogic research, collaboration for the digital platform was recurring in the sensemaking processes (Interview #10): "We have been cooperating a lot and of course that's really important since there's so many things on the neurological side that in our works are really related and close to each other. We have found really good ways to collaborate. Us speech therapists and neuropsychologists and the neurologists also".

A neurologist in the headache group also said that co-designing the digital online content for the Brain House and the digital pathway allowed "actual professionals to work together and take also customer with them" (Interview #4). In addition, the IT developer narrated that this co-design process of the platform had been easier than he expected which will help secure future funding (Interview #17):

> but I can say from this hospital district's behalf that here this is very development oriented so we want to create the services and we want to make them better and we see that this is the only way to go at the moment so I think here it's quite easy to get the funding if you can show that, if you can show good outcomes and you know, it's worth doing.

There were also expressions of "communityship" (Mintzberg, 2017) amongst stakeholders since the Health Village opened its doors. A psychiatrist from another house (for mental health) in the Health Village expressed (Interview #16):

> together with our team we just put everybody into the same pot and shook it [laughs], and wonders happen. The professionals work so well together. I think it has been completely different for example here what I've seen in the clinics before because you can't see the outfit that says I'm a doctor, I'm a nurse but you are just a person with your civil clothes and, it's very, very different and also the way that we take the patients into the conversations. So it's not only about the professional status that you have, that you compete, that okay I have this education, because then as a doctor you are learning from your patients. So then it's very different in that sense. I have seen lots of good happening in the interaction between the different professional groups in this project.

An occupational therapist from the stroke group also expressed this (Interview #6): "I am happy that everyone was equal in the team, even if hospitals are bureaucratic entities, where neurologists are generally on higher positions. I think we are really working together, as peers, which is amazing".

In interviews and in meetings observations, time and resource allocation was identified as central to co-design (Interview #14):

> I think this is great but, and our heads of the clinic, they said, oh this e-health, this is just one of the most important things in our clinic. But, you cannot see it in providing time for this. So in the evening at home I go oh, okay, let's see how. So (--) free time. And, I don't think it's right, but I haven't figured out what else to do. I want it to happen, I want it to be ready so I, I've decided, I can give some of my own time here as well of course, but…

An occupational therapist was particularly critical about support from the hospital's administration and leadership (Interview #6):

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> I asked in one of the unit meetings that when there is so much funding available, and it sounds like that most of it has gone into building a platform into that Health Village thing, so why? Isn't Finland trying to be so hi-tech that aren't such platforms already a piece of cake when you think about it as a lay person. So shouldn't they have reserved money for content development too so that people would really have some time to sit at their desk as professionals to think about what is more important for the patient in this. Somehow it feels like it is a bit unfair that some nerds receive all the money, and we are doing this content development in addition to our own work, and nobody will do anything but probably smile at some point and tell you that it's nice to have you on board. But that's how it is.

Another stakeholder also revealed the importance of administration mindset for the co-design processes (Interview #18): "well I can check the codes and everything so it's, we have, we don't need, the project itself, but we need, the administration's mindset, that they understand".

Communication of value of the diagnostics services was identified through the narratives of the co-design processes. As a former patient and a stakeholder of the stroke group recalled in their lived experiences about the co-design: "So the idea that it is a common place with all the possible information and paths and alternatives in one place is like a treasure. And then, it's just kind of the right people, like experts and doctors discussing and putting up the right things".

Another former patient described their lived experiences with diagnostic services and why it motivated them to get involved (Interview #2):

> I can't really compare the treatment I received with something else based on my own experience… I think it was everything. I cannot find anything that I think should have been done differently, so I am quite happy with it, everything went fine. But then there was this which led to these customer councils, etc., that it was such an additional bonus that is not available to everyone, not that everyone would want it, of course.

Through the interviews, stakeholders and researchers were able to notice and turn into meaning their motivation, purpose, and their contribution to the collaborative effort of co-designing a house in a village where they all belonged.

1.8. Meaning-making through mirroring the common metaphor

The final workshops were designed to mirror the metaphor of the Brain House and Health Village to stakeholders from the headache and stroke groups.

Each stakeholder was given 10 minutes to craft their stories with regards to health care, and subsequently, about the Brain House and the Health Village.

In the first workshop with the headache group, stakeholders narrated their experiences with the Health Village as transformational. The neurology leader of the group said that she could not stop thinking about the Village, even when on vacation: "I took a trip with my family to Greece and when I was looking at the people, I kept thinking "these people could benefit from our pathways too!"

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Each stakeholder in both workshops narrated their experiences with the Village as purposeful, meaningful, and transformational. For instance, an administrator, who has worked at HUS for 40 years, narrated during the first workshop that she had never experienced this type of collaboration. She reflected on how doctors traditionally kept a distance from their colleagues, and during this experience, she felt included, and her contribution was valued.

In the second workshop, the stroke group's leader, who worked closely with a stroke nurse narrated that she felt the images of digital platform to be infantile, but after co-designing the Brain House and reflecting on the metaphor, she shared that the metaphor and images gave a positive echo and helped communicate their purpose (researcher notes).

> As a first impression, I found the image naïve, like a picture from a children's book. At first, I didn't like it very much. On the other hand, I can't think what could be a better graphical presentation even though I do not like "kiiltokuva" [in Finnish something overly positive, refers to the children's old style pictures from the time prior to stickers] – like photos either, but we hope to bring health and in this sense, the name has a positive echo.

A stakeholder, who had had a stroke a few years earlier, shared that in Finnish "Health Village" comes close to a place to visit [käydä kylässä]. When there are health concerns, it is a village to visit and be welcomed in (research notes).

1.9. Action for the future through Metaphorical Tourism

Metaphorical tourism facilitates a story exchange about a common, work-related metaphor for collective sensemaking of the future (Cleland Silva and Fonseca Silva, 2022: 66-70).

In the final stage of the workshop, stakeholders Story Crafted their work on the Brain House in the Health Village. Prior to receiving coloured pencils and a blank piece of paper, they were guided by the Story Mediator with questions about co-design processes or on a collective destination. A discussion was opened as to their roles in the journey, who travels with them, responsibilities, guidelines, tools and resources needed, and the community and landscape surrounding the journey.

For the stroke group co-designing the Brain House, stakeholders agreed that the destination was a "service that working aged people who have experienced a stroke can find, the tools for their lives that match the condition to continue their lives onwards" (researcher notes).

Stakeholders shared that they were travelling with (researcher's notes):

- Ones who were experiencing a stroke and their families
- Rehabilitation house (another house in the Village)
- Service model innovation consultants

And, that their responsibilities were (researcher's notes):

- "We are responsible for our own services"
- We exchange information and know-how with the *Aivotalo* (Brain House)
- Our focus is testing our service

The picture of the Health Village was also described as (researcher notes):

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- People of different ages
- People in different situations relating to health/sickness
- Finding paths "home"

When the metaphor's image was viewed up close, stakeholders also noted that it was strange to see houses with no doors and paths that, rather than leading to the houses, went by them. This opened a discussion on accessible care and the importance for everyone in need to access their house.

This discussion was later expressed in drawings when stakeholders explored more in depth the metaphor of the Brain House in the Health Village. In the first one (Figure 4), the Brain House leader talked about her role as facilitator and communicator. She described in her drawing that more innovation over time leads to happier practitioners and patients.



Figure 4: The Brain House leader describing the metaphor where the Y axis reads "innovations" and the X axis reads "time".

In another drawing (Figure 5), the metaphor is communicated through a text along the path that read: "You will survive. You are not alone" (Finnish translation). This was described by the stakeholder as a vital purpose of the house that those in need are supported by a community of practitioners and those who care.

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Figure 5: At one end of a path, an individual holding a laptop computer. The crossroad posts on top of its display resemble the digital platform design

1.10. Noticing, meaning-making, and action of a metaphoric boundary object

Through the metaphoric boundary object of the Brain House, stakeholders were able to communicate the value of their work and their contribution to the co-design processes of the neurological diagnostics. The Brain House in the Health Village also facilitated collaboration and a boundary object emerged between the diverse communities of practice and social worlds of both stakeholders and researchers.

The metaphoric boundary object of "Brain House" and the visual avatar were initially imposed by an external IT company. However, by co-designing the "Brain House", the multi-stakeholders re-signified the semi-open boundary object through their work, their relationships with the other stakeholders and their community of practice. Our research design about the Brain House in the Health Village captured this through a year's worth of data collection and in the final two workshops.

Moreover, in the final workshops, a collective reflection on noticing and meaning-making of the co-design process of the Brain House facilitated a story exchange about what it means to visit and be welcomed to a house in a village. Or what it means to provide accessible care by collaborating with others. Through their own lived experiences of the metaphor of the Brain House and Health Village, stakeholders connected with the original purpose of the platform for collaborative accessible care. The metaphoric boundary object of the Brain House brought together the co-design process of stakeholders from multiple social worlds, including us as researchers.

Through the method of Collaborative Story Craft, the Brain House helped stakeholders communicate value for those invested in neurological diagnostics, specifically those concerned with headaches and strokes.

The Health Village emerged as the metaphoric field (Marshak, 1996), influencing how they make sense of the Brain House as a boundary object and the changes within their work practices. In Figure 6, we depict the emergence of the metaphoric boundary object in the metaphoric field.

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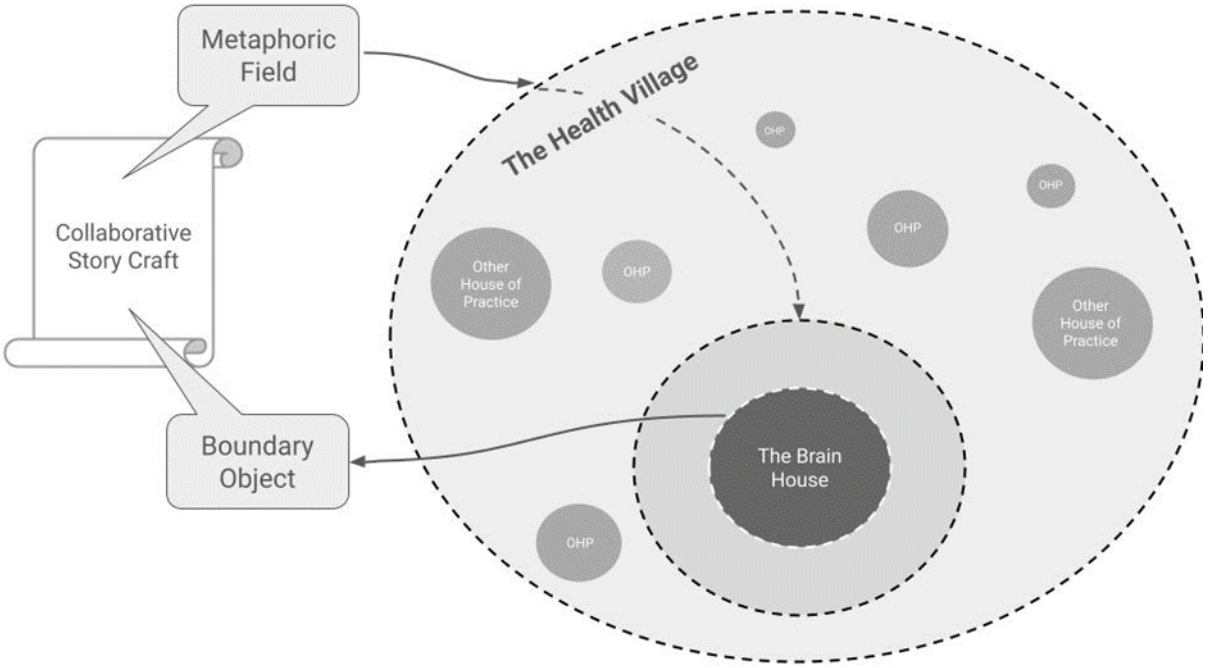


Figure 6: the Brain House as a boundary object in the Heath Village as a metaphoric field.

The model above represents the sensemaking of the co-design processes but also positions the effort of stakeholders, including researchers, using the method of Collaborative Story Craft.

5. Conclusion

In this article, we explore the co-design processes of the Brain House by stakeholders from various social worlds and communities of practice.

Through interviews, observations, and two Collaborative Story Craft workshops, we documented the collective sensemaking of co-design processes and the metaphor of the Brain House for the Health Village through the facets of noticing, meaning-making and acting.

Digital health care platforms and other e-health investments are common governmental strategies to provide accessible quality care while saving money and innovating through public-private partnership. Scholars have argued that, despite governmental optimism and investment, more studies are needed on how practitioners' work practices and interactions impact the meaning and purpose afforded to the tool.

Through our findings, we also further research recommendations to include the analysis of existing metaphors given to the tool and the significance afforded both to the metaphor and technology by multi-stakeholders. By including the existing metaphor, a deeper analysis of how the tool impacts stakeholders individually and collectively can leverage collaboration as well as the longevity of the tool and its financial investment.

The metaphor can also serve as a field to recentre the purpose, motivation, expectations, and boundaries of resources such as time to work with the tool in a sustainable and inclusive way.

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accessible care, Collaborative Story Craft, design, e-health, digital platforms, metaphor, governmental reform, public-private partnership, qualitative methods.

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Appendix 1: Interviews of stakeholders

Interview #	Occupation	Interview Date	Language
1	Neurologist	29.05.18	English
2	Retired	29.05.18	Finnish
3	Stroke nurse	31.5.18	Finnish
4	Neurologist	1.6.18	English
5	Government employee	1.6.18	Finnish
6	Physiotherapist	6.6.18	Finnish
7	Psychiatric Director	8.6.18	Finnish
8	Primary care physician	8.6.18	Finnish
9	Neurologist	12.6.18	Finnish
10	Speech therapist	8.8.18	English
11	Psychiatrist and IT Development line for psychosocial treatments	12.6.18	Finnish
12	Stroke nurse	14.6.18	Finnish
13	IT project developer	15.6.18	Finnish
14	Neurologist	19.6.18	Finnish
15	Neurologist	21.6.18	Finnish
16	Psychiatrist and Head of ICT psychiatry and psychosocial line	22.6.18	Finnish
17	Midwife and Health Village business developer	25.6.18	English
18	E-service coordinator and physiotherapist	20.08.18	English

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Appendix 2: Interview questions for health care practitioners at HUS*

1. What is your relationship with HUS [Helsinki University Hospital] and the [Stroke or Headache] group? If you have a position at HUS, what is it?
2. During a working day, could you describe your work and responsibilities at HUS?
3. Could you tell us about your experiences and education as a health care professional or as a professional working in health care?
4. From your point of view, how would you describe SOTE [social and health care governmental reform towards Private Public Partnerships in Finland]?
5. What are you views on the digitalization of health care services?
6. Have you participated in digitalization of health care services at HUS? If so, how?
7. How has SOTE affected your work or how do you interact with care services?
8. Has digitalization affected your work? If so, how?
9. Have your received resources and support to facilitate requests from SOTE?
10. Do you feel that you have access to tools, resources, and support to digitalize the health care service you provide to patients?
11. Do you believe that SOTE impacts your work and your relationships with the patient?
12. Is there anything else you would like to discuss or share about SOTE and digitalised health care services?

The questions to the patients and patients' organizations were similar but related to the co-design of the digital services and receiving care as for stroke or headache.

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